

ISO 226 and Equal-Loudness Curves (ELC)

Equal loudness curves were first described and plotted by Harvey Fletcher and Wilden Munson in the 1930s. Like many other influential acoustic researchers of their time, they worked together at Bell Labs. In order to measure relative perceptions of loudness across varying frequencies, they gathered experimental data of human perceptions in a controlled laboratory setting that allowed normalization of as many other variables as possible. This required using well-calibrated equipment, over-ear headphones, and an anechoic testing environment.

For a full spectral range of frequencies, they compared the relative loudness perception of pure sine wave tones. They used the standard reference still in place today, that the minimum threshold of perception for a 1kHz tone (1000 cycles per second) is 0 decibels (dB), and labeled contour curves of the same perceived loudness for 10 dB intervals up to 120 dB. Their chart looks much like a topographic map for sound loudness perception. They found that the lowest frequency/longest wavelength tones and the highest frequency/shortest wavelength tones had to be played at higher amplitudes to be perceived as the same loudness as middle-frequency tones. They also reported the phenomenon of increased sensitivity between 3-4 kHz, considered to be a characteristic of resonance in human ear anatomy.

The curves reported by Fletcher and Munson have been revised every couple of decades, as scientists have refined their understanding of psychoacoustics and human audition. Today the International Organization for Standardization (ISO) maintains official global standards for ELC, called “ISO 226: Acoustics — Normal Equal-Loudness-Level Contours.” The most recent version of this standard was published in 2023, and the previous revision was in 2003. The overview of ISO 226 is publicly available, but the full document must be purchased to view. However, there are secondary resources, such as the PyDSM library for Python, that do provide open access to contour data.

According to a note in the introduction of the standard, the pure tone curves are less applicable as broader bands of sound are considered. There is insufficient data available to plot similar standards for sounds consisting of more complex spectral envelopes. Also, the ISO 226 standard includes only data for typical stereo hearing from young adults ages 18 to 25.

Audio engineers can use the ISO 226 data in a variety of applications. It can help to balance music and other audio mixing, ensuring future listeners experience the full spectrum of recordings as intended. The ELC values could also be used to balance the relative volume of dialog and music in film and TV soundtracks. Regardless of the context, it is useful to understand not only the principles behind this standard as a tool, but also their limitations. For example, audio engineers should consider how use of these curves will impact the experience of listeners with hearing differences, whether due to disability, ordinary aging progression, or even temporary environmental factors. The principles behind ELC are helpful to understand for anyone working with sound. They also demonstrate a key overarching concept in audio that perception of sound is nonlinear and varies by individual.

Sources:

Cramer, William S. (June 1983). "Wilden A. Munson 1902-1982," Acoustical News—USA, Acoustical Society of America. Retrieved Sept. 12, 2026 from <https://pubs.aip.org/asa/jasa/article-abstract/73/6/2232/628316/Munson-Wilden-A-1902-1982>

Georgescu I. (2022). Bringing back the golden days of Bell Labs. Nature reviews. Physics, 4(2), 76–78. Retrieved Sept. 12, 2026 from <https://doi.org/10.1038/s42254-022-00426-6>

International Organization for Standardization. (March 2023). ISO 226:2023 Acoustics — Normal equal-loudness-level contours. Retrieved publicly available sections Sept. 12, 2026 from <https://www.iso.org/standard/83117.html>

Rabiner, Lawrence. (May 2004). "Digital signal processing at Bell Labs—Foundations for speech and acoustics research" (meeting abstract). Journal of the Acoustical Society of America, vol. 115 (5). Retrieved Sept. 12, 2026 from <https://doi.org/10.1121/1.4782758>

Suzuki Y, Takeshima H, and Kurakata K. (October 2023). Revision of ISA 226 "Normal Equal-Loudness-Level Countours" from 2003 to 2023 edition: The background and results. Acoustical Science and Technology, vol. 45(1). Retrieved Sept. 12, 2026 from <https://doi.org/10.1250/ast.e23.66>

Titze, Ingo R. "Hall of Fame, Harvey Fletcher, Voice and Speech Science," The National Center for Voice and Speech (NCVS). Retrieved Sept. 12, 2026 from <https://ncvs.org/hall-of-fame/hall-of-fame-harvey-fletcher/>